

9.1 What Is Network Management?

Before diving in to network management itself, let's first consider a few illustrative “real-world” non-networking scenarios in which a complex system with many interacting components must be monitored, managed, and controlled by an administrator. Electrical power-generation plants have a control room where dials, gauges, and lights monitor the status (temperature, pressure, flow) of remote valves, pipes, vessels, and other plant components. These devices allow the operator to monitor the plant's many components, and may alert the operator (with the famous flashing red warning light) when trouble is imminent. Actions are taken by the plant operator to control these components. Similarly, an airplane cockpit is instrumented to allow a pilot to monitor and control the many components that make up an airplane. In these two examples, the “administrator” *monitors* remote devices and *analyzes* their data to ensure that they are operational and operating within prescribed limits (for example, that a core meltdown of a nuclear power plant is not imminent, or that the plane is not about to run out of fuel), *reactively controls* the system by making adjustments in response to the changes within the system or its environment, and *proactively manages* the system (for example, by detecting trends or anomalous behavior, allowing action to be taken before serious problems arise). In a similar sense, the network administrator will actively monitor, manage, and control the system with which she or he is entrusted.

In the early days of networking, when computer networks were research artifacts rather than a critical infrastructure used by hundreds of millions of people a day, “network management” was unheard of. If one encountered a network problem, one might run a few pings to locate the source of the problem and then modify system settings, reboot hardware or software, or call a remote colleague to do so. (A very readable discussion of the first major “crash” of the ARPAnet on October 27, 1980, long before network management tools were available, and the efforts taken to recover from and understand the crash is [RFC 789].) As the public Internet and private intranets have grown from small networks into a large global infrastructure, the need to manage the huge number of hardware and software components within these networks more systematically has grown more important as well.

In order to motivate our study of network management, let's begin with a simple example. Figure 9.1 illustrates a small network consisting of three routers and a number of hosts and servers. Even in such a simple network, there are many scenarios in which a network administrator might benefit tremendously from having appropriate network management tools:

- *Detecting failure of an interface card at a host or a router.* With appropriate network management tools, a network entity (for example, router A) may report to the network administrator that one of its interfaces has gone down. (This is certainly preferable to a phone call to the NOC from an irate user who says the network connection is down!) A network administrator who actively monitors

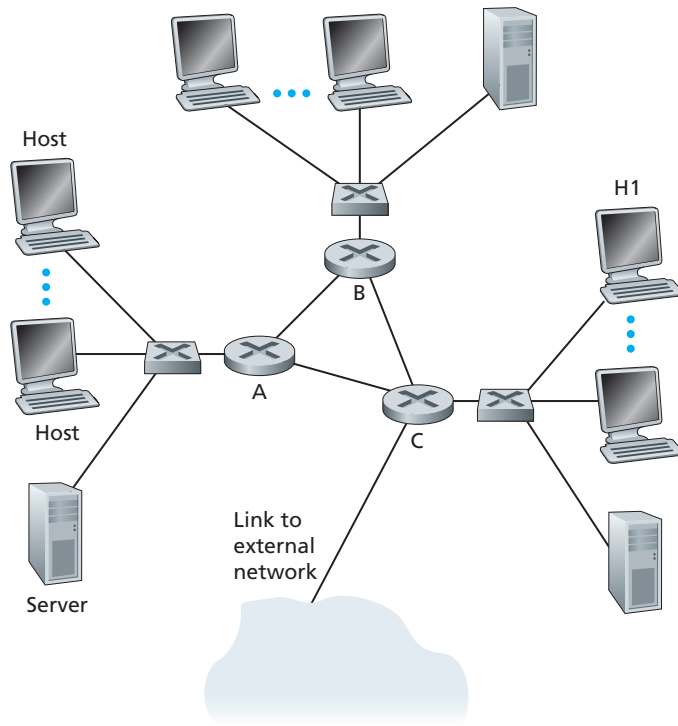


Figure 9.1 ♦ A simple scenario illustrating the uses of network management

and analyzes network traffic may be able to *really* impress the would-be irate user by detecting problems in the interface ahead of time and replacing the interface card before it fails. This might be done, for example, if the administrator noted an increase in checksum errors in frames being sent by the soon-to-die interface.

- *Host monitoring.* Here, the network administrator might periodically check to see if all network hosts are up and operational. Once again, the network administrator may really be able to impress a network user by proactively responding to a problem (host down) before it is reported by a user.
- *Monitoring traffic to aid in resource deployment.* A network administrator might monitor source-to-destination traffic patterns and notice, for example, that by switching servers between LAN segments, the amount of traffic that crosses multiple LANs could be significantly decreased. Imagine the happiness all around when better performance is achieved with no new equipment costs. Similarly, by monitoring link utilization, a network administrator might determine that a LAN segment or the external link to the outside world is overloaded and

that a higher-bandwidth link should thus be provisioned (alas, at an increased cost). The network administrator might also want to be notified automatically when congestion levels on a link exceed a given threshold value, in order to provision a higher-bandwidth link before congestion becomes serious.

- *Detecting rapid changes in routing tables.* Route flapping—frequent changes in the routing tables—may indicate instabilities in the routing or a misconfigured router. Certainly, the network administrator who has improperly configured a router would prefer to discover the error him- or herself, before the network goes down.
- *Monitoring for SLAs.* **Service Level Agreements (SLAs)** are contracts that define specific performance metrics and acceptable levels of network-provider performance with respect to these metrics [Huston 1999a]. Verizon and Sprint are just two of the many network providers that guarantee SLAs [AT&T SLA 2012; Verizon SLA 2012] to their customers. These SLAs include service availability (outage), latency, throughput, and outage notification requirements. Clearly, if performance criteria are to be part of a service agreement between a network provider and its users, then measuring and managing performance will be of great importance to the network administrator.
- *Intrusion detection.* A network administrator may want to be notified when network traffic arrives from, or is destined for, a suspicious source (for example, host or port number). Similarly, a network administrator may want to detect (and in many cases filter) the existence of certain types of traffic (for example, source-routed packets, or a large number of SYN packets directed to a given host) that are known to be characteristic of the types of security attacks that we considered in Chapter 8.

The International Organization for Standardization (ISO) has created a network management model that is useful for placing the anecdotal scenarios above in a more structured framework. Five areas of network management are defined:

- *Performance management.* The goal of performance management is to quantify, measure, report, analyze, and control the performance (for example, utilization and throughput) of different network components. These components include individual devices (for example, links, routers, and hosts) as well as end-to-end abstractions such as a path through the network. We will see shortly that protocol standards such as the Simple Network Management Protocol (SNMP) [RFC 3410] play a central role in Internet performance management.
- *Fault management.* The goal of fault management is to log, detect, and respond to fault conditions in the network. The line between fault management and performance management is rather blurred. We can think of fault management as the immediate handling of transient network failures (for example, link, host, or router hardware or software outages), while performance management takes

the longer-term view of providing acceptable levels of performance in the face of varying traffic demands and occasional network device failures. As with performance management, the SNMP protocol plays a central role in fault management.

- *Configuration management.* Configuration management allows a network manager to track which devices are on the managed network and the hardware and software configurations of these devices. An overview of configuration management and requirements for IP-based networks can be found in [RFC 3139].
- *Accounting management.* Accounting management allows the network manager to specify, log, and control user and device access to network resources. Usage quotas, usage-based charging, and the allocation of resource-access privileges all fall under accounting management.
- *Security management.* The goal of security management is to control access to network resources according to some well-defined policy. The key distribution centers that we studied in Section 8.3 are components of security management. The use of firewalls to monitor and control external access points to one's network, a topic we studied in Section 8.9, is another crucial component.

In this chapter, we'll cover only the rudiments of network management. Our focus will be purposefully narrow—we'll examine only the *infrastructure* for network management—the overall architecture, network management protocols, and information base through which a network administrator keeps the network up and running. We'll *not* cover the decision-making processes of the network administrator, who must plan, analyze, and respond to the management information that is conveyed to the NOC. In this area, topics such as fault identification and management [Katzela 1995; Medhi 1997; Labovitz 1997; Steinder 2002; Feamster 2005; Wu 2005; Teixeira 2006], anomaly detection [Lakhina 2004; Lakhina 2005; Barford 2009], and more come into consideration. Nor will we cover the broader topic of service management [Saydam 1996; RFC 3052]—the provisioning of resources such as bandwidth, server capacity, and the other computational/communication resources needed to meet the mission-specific service requirements of an enterprise.

An often-asked question is “What is network management?” Our discussion above has motivated the need for, and illustrated a few of the uses of, network management. We'll conclude this section with a single-sentence (albeit a rather long run-on sentence) definition of network management from [Saydam 1996]:

“Network management includes the deployment, integration, and coordination of the hardware, software, and human elements to monitor, test, poll, configure, analyze, evaluate, and control the network and element resources to meet the real-time, operational performance, and Quality of Service requirements at a reasonable cost.”

It's a mouthful, but it's a good workable definition. In the following sections, we'll add some meat to this rather bare-bones definition of network management.

9.2 The Infrastructure for Network Management

We've seen in the preceding section that network management requires the ability to “monitor, test, poll, configure, . . . and control” the hardware and software components in a network. Because the network devices are distributed, this will, at a minimum, require that the network administrator be able to gather data (for example, for monitoring purposes) from a remote entity and effect changes at that remote entity (for example, control it). A human analogy will prove useful here for understanding the infrastructure needed for network management.

Imagine that you're the head of a large organization that has branch offices around the world. It's your job to make sure that the pieces of your organization are operating smoothly. How will you do so? At a minimum, you'll periodically gather data from your branch offices in the form of reports and various quantitative measures of activity, productivity, and budget. You'll occasionally (but not always) be explicitly notified when there's a problem in one of the branch offices; the branch manager who wants to climb the corporate ladder (perhaps to get your job) may send you unsolicited reports indicating how smoothly things are running at his or her branch. You'll sift through the reports you receive, hoping to find smooth operations everywhere but no doubt finding problems in need of your attention. You might initiate a one-on-one dialogue with one of your problem branch offices, gather more data in order to understand the problem, and then pass down an executive order (“Make this change!”) to the branch office manager.

Implicit in this very common human scenario is an infrastructure for controlling the organization—the boss (you), the remote sites being controlled (the branch offices), your remote agents (the branch office managers), communication protocols (for transmitting standard reports and data, and for one-on-one dialogues), and data (the report contents and the quantitative measures of activity, productivity, and budget). Each of these components in human organizational management has a counterpart in network management.

The architecture of a network management system is conceptually identical to this simple human organizational analogy. The network management field has its own specific terminology for the various components of a network management architecture, and so we adopt that terminology here. As shown in Figure 9.2, there are three principal components of a network management architecture: a managing entity (the boss in our analogy above—you), the managed devices (the branch office), and a network management protocol.